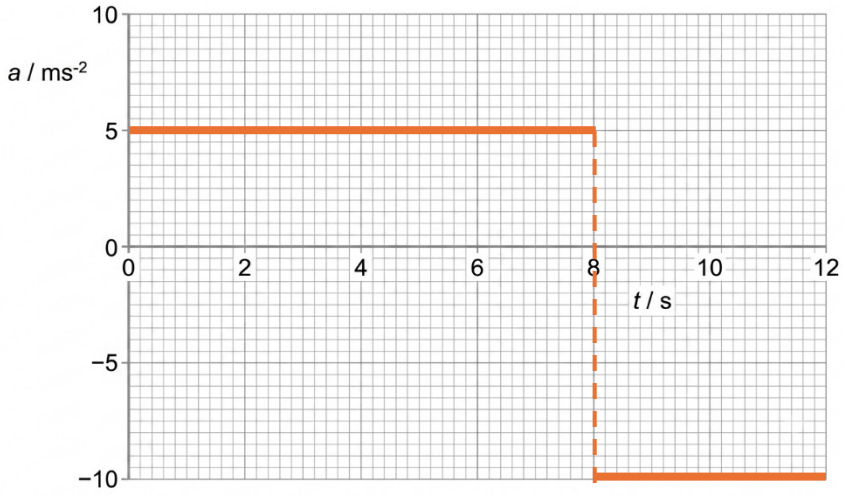
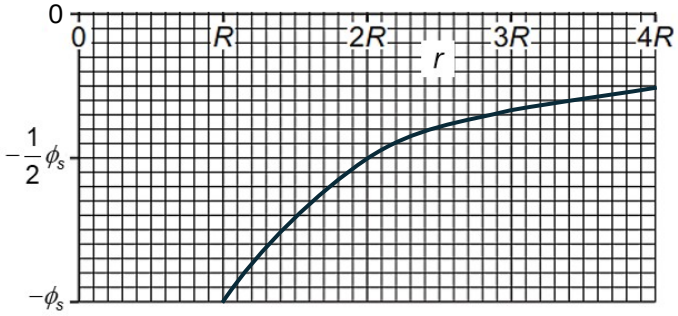
1	**(a)**	Maximum momentum = 7.2×10^4 N s Maximum KE = $\frac{p^2}{2m} = \frac{(7.2\times10^4)^2}{2\times1800} = 1.4\times10^6$ J	**C1** **A1**
	(b)	*v*-*t* graph follows the same shape with maximum speed of 40 m s^{-1} Area under *v*-*t* graph = $0.5 \times 40 \times 12$ = 240 m	**C1** **A1**
	(c)	 Gradient of *p*-*t* graph = force, Acceleration = force/mass Horizontal line from (0,5.0) to (8.0, 5.0) and horizontal line from (8.0,-10.0) to (12.0,-10.0)	 **B1** **B1**

2	**(a)**	$s_Y = u_Y t + \frac{1}{2}a_Y t^2$ $80 = 0 + \frac{1}{2}(9.81)t^2$ t = 4.0 s	**C1** **A1**
	(b)	Horizontal distance = 25×4.0 = 100 m	**A1**
	(c)	Distance covered by hiker = 4.0×4.0 = 16 m Hiker is at a position of 56 m (40 m + 16 m) from the start point and the package lands at a position of 100 m. Thus, the hiker does not catch the package.	**M1** **A1**

3	**(a)(i)**	curve with continuously decreasing magnitude and with gradient of continuously decreasing magnitude, starting at (R, $-\phi_s$) line passing through ($2R$, $-\frac{1}{2}\phi_s$) and ($4R$, $-\frac{1}{4}\phi_s$)  	**B1**
	(a)(ii)	Gravitational potential at infinity is taken to be zero. OR An external force has to be applied in opposite direction to gravitational force. (opposite to the displacement) Work done to bring a mass from infinity to the point in the field is negative.	**B1** **B1**
	(b)	$\phi = -\frac{GM}{r}$ GM = gradient of the graph $M = \frac{1.76 \times 10^{8}}{3.0 \times 10^{-8} \times 6.67 \times 10^{-11}}$ $M = 8.8 \times 10^{25}$ kg Accept if student just use one point on the graph.	**M1** **A0**
	(c)(i)	Gravitational force provides the centripetal force for the circular motion. $F_G = F_C$ $\frac{GMm}{r^2} = mr(\frac{2\pi}{T})^2$ $\frac{(6.67 \times 10^{-11})(8.8 \times 10^{25})}{r^2} = r(\frac{2\pi}{0.72 \times 24 \times 60 \times 60})^2$ $r = 8.3 \times 10^{7}$ m If student did not convert the 0.72 Earth days to seconds, deduct 1 mark for this question.	**B1** **C1** **A1**
	(c)(ii)	Using principle of conservation of energy $GPE_i + KE_i$ + Additional energy = $GPE_f + KE_f$ $GPE_i + KE_i$ + Additional energy = 0	

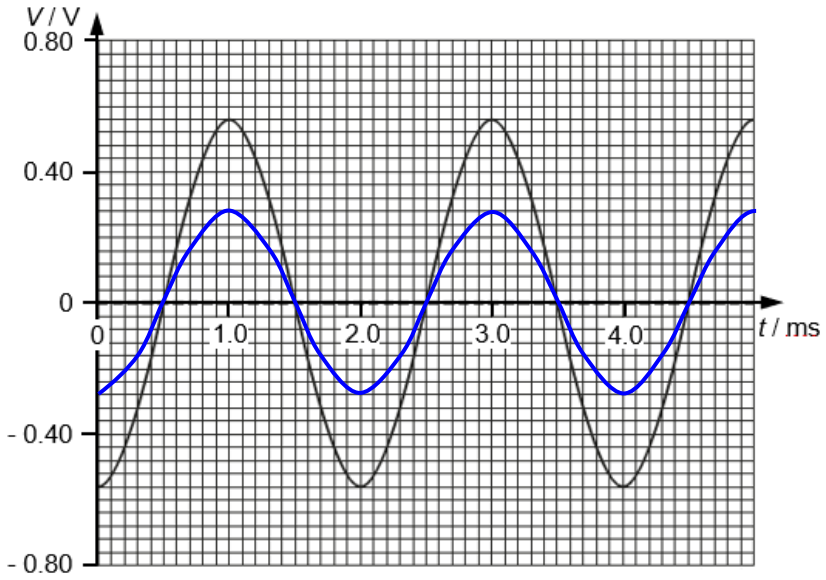
		$-\frac{GMm}{r}+(\frac{GMm}{2r})$ + Additional energy = 0 Additional energy = $\frac{GMm}{2r}$ $=\frac{(6.67\times10^{-11})(8.8\times10^{25})(1200)}{2(8.3\times10^{7})}$ $=4.24\times10^{10}$ J	**C1** **C1** **A1**

4	**(a)(i)**	The volume decreases, the <u>time between collisions with wall of cylinder is reduced</u> / The frequency of collision with the walls of the cylinder increases. The <u>average force on the wall / rate of change of momentum of molecules is increased.</u> Hence, pressure increases.	**B1** **B1**
	(a)(ii)	1. $V \alpha T$ Thus, $V_2 = V_1(T_2/T_1)$ $= (5.0 \times 10^{-3})\ (80 + 273.15)/(35 + 273.15)$ $= 5.7 \times 10^{-3}$ m^3	**C1** **A1**
		2. $W = p\ \Delta V$ $= (1.0 \times 10^{5})(5.7 \times 10^{-3} - 5.0 \times 10^{-3})$ = 70 J	**A1**
	(b)(i)	P = 230 (8) = 1840 W	**A1**
	(b)(ii)	$Q = m\ l_v$ 1840 (10) = $(8.1 \times 10^{-3})\ l_v$ $l_v = 2.27 \times 10^{6}$ J kg^{-1}	**C1** **A1**
	(b)(iii)	<u>Water vapour condensing at the inner lid</u> of the kettle and <u>dripping back</u> into the water. The <u>decrease in the reading of the balance is less</u> which causes the calculated value of l_v to be <u>greater</u> / an <u>overestimation</u>.	**B1** **B1**

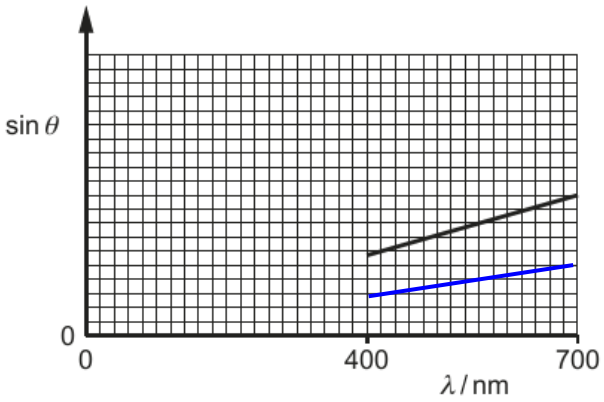
5	**(a)**	Electric field strength at a point in an electric field is defined as the <u>electric force per unit charge</u> on a <u>positive charge</u> placed at that point. [1]	**B1**
	(b)(i)	$E = V/d$ = 4.5 / (0.030) = 150 V m^{-1}	**C1** **A1**
	(b)(ii)	Loss in KE = gain in EPE or $KE_B + EPE_B = KE_A + EPE_A$ $\frac{1}{2} m u^2 = e V$ or $\frac{1}{2} m u^2 + (-1.6 \times 10^{-19})(0) = 0 + (-1.6 \times 10^{-19})(-4.5)$ $u = \sqrt{\frac{2(1.6\times10^{-19})(4.5)}{9.11\times10^{-31}}}$ [1] $=1.26\times10^{6}$ m s^{-1} [1] Alternative (Kinematics): $a = eE/m$ $v^2 = u^2 + 2as$ $0^2 = u^2 + 2\left[\frac{(-1.6\times10^{-19})(150)}{9.11\times10^{-31}}\right](30\times10^{-3})$ u = 1.26 $\times$ 10^6 m s^{-1}	**C1** **A1**
	(b)(iii)	The minimum value of u in order for electrons to reach plate A is <u>dependent on the potential difference between the plates</u>. [1] Thus the value in (b)(ii) is <u>unchanged.</u> [1] If student states that it is independent of distance and that the value remains unchanged, he will get 1 mark for the A1.	**M1** **A1**
	(c)(i)	$Q = CV$ = (160 $\times$ 10^{-6})(200) = 0.032 C [1]	**A1**
	(c)(ii)	$\Delta E = \frac{1}{2} C (V_2^2 - V_1^2) = \frac{1}{2}$ (160 $\times$ 10^{-6})(200^2 - 100^2) [1] = 2.4 J [1]	**C1** **A1**

6	**(a)**	The magnetic force acting on the electrons provides its centripetal force. $F_c = ma_c$ $Bqv = \frac{mv^2}{r}$ $8.0 \times 10^{-4} \times 1.60 \times 10^{-19} = \frac{9.11 \times 10^{-31} \times v}{6.4 \times 10^{-2}}$ $v = 8.992 \times 10^6 \approx 9.0 \times 10^6 \text{ ms}^{-1}$	**C1** **A1**
	(b)(i)	For the electron to pass through undeviated, the net force acting on it must be zero. This requires the electric force and the magnetic force to be equal in magnitude but opposite in direction. Since the magnetic force on the electron is directed downwards, the electric force must act upwards to balance it. Since an electric force acting on a negatively charged particle is in the opposite direction to that of the electric field lines, the electric field must be directed downwards.	**A1**
	(b)(ii)	$F_E = F_B$ $qE = Bqv$ $q \times E = 8.0 \times 10^{-4} \times q \times 8.992 \times 10^6$ $E = 7.193 \times 10^3 \approx 7.2 \times 10^3 \text{ NC}^{-1}$	**A1**
	(c)	The magnetic force acting on the positron is now directed upwards while the electric force acting on the positron is now directed downwards. The resultant force acting on the positron remains unchanged. Hence the positron continues to move in a straight line or undeviated.	**B1** **B1**

7	**(a)(i)**	The magnitude of the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage.	B1 B1
	(a)(ii)	The direction of the induced e.m.f. produced causes an effect to oppose the change that produces it.	B1 B1
	(b)	As magnet oscillates, there is change in magnetic flux linkage in the coil. This causes an induced e.m.f. , which drives a current in the coil and resistor (since switch is closed). The current in the coil produces a force opposing motion of the magnet. The force does negative work / There is damping in the oscillation / Mechanical energy is transformed to electrical energy and then dissipated as thermal energy in the resistor. Thus, the magnet's amplitude decreases.	B1 B1 B1 B1

8	**(a)(i)**	$T = 4 \times 0.50 \times 10^{-3}$ $T = 2.0 \times 10^{-3}$ s $f = \dfrac{1}{2.0 \times 10^{-3}}$ $f = 500$ Hz	**A1**
	(a)(ii)	 Intensity of the wave is proportional to the square of its amplitude. Since the intensity is reduced to one quarter, the amplitude is reduced to half. The period of the wave remains unchanged.	**B1** **B1**
	(b)(i)	The incident sound wave (travels towards the metal sheet and) reflects at the metal sheet, creating a reflected wave that travels in opposite direction. Since both waves possess the same frequency and amplitude, they continuously superpose as they pass through each other, creating a stationary wave.	**B1** **B1** **A0**
	(b)(ii)	$d = 1.5\lambda$ $1.05 = 1.5\lambda$ $\lambda = 0.70$ m $v = f\lambda$ $v = 500 \times 0.70$ $v = 350$ m s^{-1}	**C1** **C1** **A1**
	(b)(iii)	Since the material absorb sound, the reflected wave has a smaller amplitude than the incident wave. Hence there is incomplete cancellation at the nodes.	**M1**

		The amplitude registered at the minimum position increases. BOD if students state that the amplitude is "no longer zero"	**A1**
	(c)(i)1.	Since point Q is at the first order maximum. phase difference = $1 \times 2\pi = 360°$ (accept 0°)	**A1**
	(c)(i)2.	Since point P is at the second order minimum, path difference = $1.5\lambda = 1.5 \times 610$ $= 915 \approx 920$ nm	**A1**
	(c)(ii)	$x = \frac{1}{4} \times 22 \times 10^{-3} = 5.5 \times 10^{-3}$ $\frac{ax}{D} = \lambda$ $\frac{a \times 5.5 \times 10^{-3}}{2.7} = 610 \times 10^{-9}$ $a = 2.995 \times 10^{-4} \approx 3.0 \times 10^{-4}$ m	**C1** **A1**
	(c)(iii)	All component wavelengths of white light have a path difference of zero at the center, causing them to superpose completely constructively. The central bright fringe appears as a (sharp) white line. Since fringe separation is directly proportional to the wavelength, The side fringes on either side of this center fringe appear as continuous colored spectra where violet is closest to the center and red is furthest away.	**B1** **B1**
	(d)(i)	$G = \frac{\sin\theta}{\lambda}$ $a\sin\theta = n\lambda$ $d = n\frac{\lambda}{\sin\theta}$ $d = \frac{4}{G}$	**C1** **A1**

	(d)(ii)	 $a\sin\theta = n\lambda$ $\frac{\sin\theta}{\lambda} = \frac{n}{a}$ Since the order of the maximum is reduced from 4 to 2, the gradient of the line should be halved. straight line from 400 nm to 700 nm that is always below the printed line straight line has half the gradient as the printed line.	**M1** **A1**

9	**(a)(i)**	$m = 235 \times 1.66 \times 10^{-27} = 3.90 \times 10^{-25}$ kg	A1
	(a)(ii)	$v = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times (8.9 \times 10^{-15})^3$ Volume = 2.95×10^{-42} m^3 Density = $\frac{m}{v} = \frac{3.90 \times 10^{-25}}{2.95 \times 10^{-42}} = 1.32 \times 10^{17}$	B1 A1
	(a)(iii)	Since the density of the lump of uranium is a very small fraction of the density of the nucleus, (need to show comparison) this means that there are lots of empty space in the atom. / volume of <u>atom</u> is very large compared to volume of <u>nucleus</u> / mass is concentrated in a very small region (nucleus).	B1 B1
	(b)(i)	A nuclear reaction in which a <u>heavy nucleus splits</u> into two or more lighter nuclei, with release of energy.	B1 B1
	(b)(ii)	**1.** Binding energy of product particles = $144 \times 1.3341 \times 10^{-12} + 90 \times 1.3864 \times 10^{-12}$ = 3.169×10^{-10} J Binding energy of reactant particles = $235 \times 1.2191 \times 10^{-12}$ = 2.865×10^{-10} J Energy released = 3.169×10^{-10} - 2.865×10^{-10} = 3.04×10^{-11} J	B1 B1 A1
	(b)(ii)	$\frac{\text{energy released}}{c^2} = \frac{3.04 \times 10^{-11}}{(3.00 \times 10^8)^2}$ **2.** mass equivalent = = 3.38×10^{-28} kg	B1 A1
	(b)(iii)	(The neutrons are constituent particles/free unbounded/single particles,) hence <u>there is no binding energy for the neutrons.</u>	B1

	(c)(i)	Decay constant is probability per unit time of decay of a radioactive nucleus.	**A1**
	(c)(ii)1.	$\lambda = \frac{\ln 2}{28.0 \times 365 \times 24 \times 60 \times 60} = 7.85 \times 10^{-10}$ s^{-1}	**A1**
	(c)(ii)2.	$A = \lambda N \Rightarrow N = \frac{6.4 \times 10^{9}}{7.8 \times 10^{-10}} = 8.21 \times 10^{18}$ mass of sample = $8.21 \times 10^{18} \times 90 \times 1.66 \times 10^{-27}$ = 1.23×10^{-6} kg = 1.2×10^{-3} g	**C1** **C1** **A1**
	(c)(ii)3.	volume = $\frac{1.23 \times 10^{-3}}{2.54} = 4.8 \times 10^{-4}$ cm^3	**A1**
	(c)(iii)	The volume calculated in **(c)(ii)3.** is a very small volume of Strontium-90. It shows that this small volume of Strontium-90 already has high activity (10^9 Bq) compared to background count (10^{-1} Bq). Inhaling in dust particles exposes one to high radiation.	**B1**